

Computer Vision Engineer — Technical Task

iCloudEMS · Smart Campus Team · Duration: 90 minutes

Context

iCloudEMS is building campus intelligence on top of CCTV cameras that already exist on campus — no new hardware, no wristbands, no apps. The end goal: recognise students for automatic attendance, flag overcrowded or empty rooms, and catch wasted utilities like lights left on in an empty room. This task tests a small, realistic slice of that pipeline on a live video stream, not a saved file.

What You Need Before You Start

This task runs on a live stream, not a file you process all at once. You have two options for the feed:

- Point your laptop webcam at yourself or a room and act out people entering, leaving, sitting, and moving around.
- Download any suitable video (stock sites, YouTube, public entrance/classroom datasets) and stream it in real time — read and process it frame by frame as it plays, not by loading the whole file and analysing it as a batch.

Either way, the video should show people entering/exiting a room, some sitting, some standing or walking.

The Task

Build a pipeline that processes your live stream and outputs the following, updated continuously as the stream plays:

What to detect	Example output
Person enters vs. exits the room	"Entry detected" / "Exit detected"
Motion in the frame	"Motion detected" / "No motion"
Running attendance — currently present	"5 currently present"
Total unique people who entered so far	"8 unique entries"
How many are seated vs. standing/moving	"Seated: 4, Standing/Moving: 2"
Frame quality — clear or blurry	"Clear" / "Blurry — flag for review"
Room occupied or empty right now	"Occupied" / "Empty"

Bonus, if time allows: flag a moment where the room shows lights on but nobody currently present.

Near the end of the session, you will run your pipeline live on a new stream you haven't tested against before — either your webcam on the spot, or a new clip streamed in real time.

Rules — What's Allowed

- Use any AI coding assistant you want (Copilot, Cursor, ChatGPT, Claude, etc.) — this is fully open-book.
- Use any pretrained detection or tracking model/library (YOLO, MediaPipe, OpenCV, DeepSORT, ByteTrack, etc.). You are not expected to train anything from scratch.
- Full internet access.
- Processing must happen on the stream as it arrives, frame by frame — batch-analysing a fully-loaded file does not meet the bar.
- Processing every single frame is not required — sampling at a reasonable rate (e.g. 5–10 fps) is fine and often the smarter choice.
- You must run it yourself, on your own machine, and demo it live — not just paste code.
- A simple UI or output view is required (Streamlit, Gradio, an annotated live window, or a live-updating dashboard/console). Silent scripts with no visible output are not enough.

Time Plan (90 minutes)

Time	What happens
0:00 – 1:15	Build your pipeline and test it against your own stream
1:15 – 1:30	Live demo — run it on a new stream you haven't tested against before

What We're Actually Evaluating

The code itself is not the point — plenty of people can wire up a YOLO model with an AI assistant's help. We're evaluating:

- **Tool judgment** — do you pick a sensible detection + tracking approach quickly, or thrash between options?
- **Real-time performance** — does your pipeline keep up with a live stream without major lag or freezing?
- **Handling real movement** — how do you deal with someone walking in and back out, two people crossing at once, or someone lingering in the doorway?
- **Debugging under pressure** — when it breaks on the new stream, how do you reason through it?
- **Systems thinking** — do your design choices show awareness of edge vs. cloud processing, latency, and failure cases at scale?
- **Honesty** — can you state plainly what your solution won't handle (e.g. "this miscounts if two people overlap") instead of overselling it?

Include With Your Submission

Write brief answers (2–4 sentences each) to the following and submit them alongside your code:

- How would you scale this from one live camera to 500 cameras streaming at once? Where would it break first?
 - How would you avoid double-counting or losing track of a person if they briefly leave the camera's view?
 - How would you handle a camera feed that's consistently blurry or poor quality — flag it, skip it, or something else?
-

